

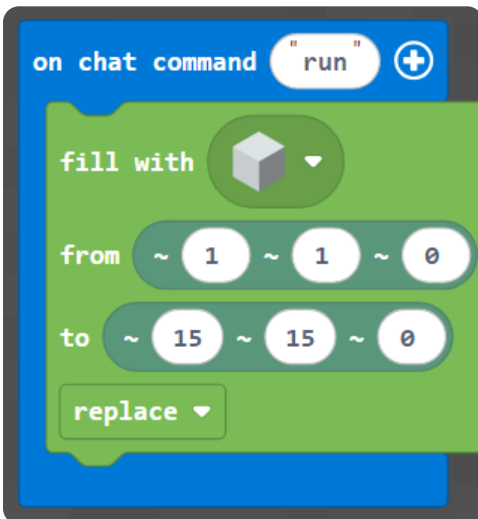
M003 Extension — English Worksheet (Beginner)

Topic: Pixel-Art Snow Golem (x, y) · **Course:** 3D Coordinates · **Level:** Extension (Beginner, after Week 3) · **Time:** about 30 minutes

A bonus picture challenge: build a **snow golem** on your wall using (x, y) — x is **across**, y is **up**. Read the picture and place each block at its (x, y).

Build your canvas. Run this to make a blank 15 × 15 wall, then put a **red block at your feet** as your home spot:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too.

1 · Predict

Read the steps. Circle your answer.

```
place pumpkin block at (6, 15)
place pumpkin block at (7, 15)
place pumpkin block at (8, 15)
```

A line across or up? Circle one: across · up

Which number stays the same? Circle one: x · y

```
place snow block at (4, 2)
place snow block at (4, 3)
place snow block at (4, 4)
```

Across or up? Circle one: across · up

Which number changes? Circle one: x · y

2 · Spot the Bug 🐛

This code is broken. Read what it should do, then fill in the fix.

Bug — The golem's two eyes should be at **two different spots**: (7, 13) and (9, 13). Both blocks land in the **same spot**.

```
place black block at (7, 13)
place black block at (7, 13)
```

Hint: the first number (x) of the second eye is wrong. One eye is further **across** than the other, so its x must be **bigger**.

Fill the blank to fix the second line:

```
place black block at (7, 13)
place black block at (__ , 13)
```

Which number fills the blank? Circle one: 7 · 9

Why was the code wrong? Circle one: both eyes had the same x · the color was wrong

3 · Show Your Work 📷 🎥

Stand on your **home spot** (the red block).

Warm up first. Copy this small pumpkin head, reading each square's (x, y).

🎃 **Pumpkin head (orange · #e5772b, face · black)**

	1	2	3	4	5
5	orange	orange	orange	orange	orange
4	orange	black	orange	black	orange
3	orange	orange	orange	orange	orange
2	orange	black	black	black	orange
1	orange	orange	orange	orange	orange

Try the big one. Copy the whole 🧊 snow golem below, reading each colored square's (x, y).

🧊 **Snow Golem (snow · light blue #cfe0ec, pumpkin · #e5772b, face + coal buttons · black, arms · brown #6b4a2b)**

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15						orange	orange	orange	orange	orange					
14						orange	orange	orange	orange	orange					
13						orange	black	orange	black	orange					
12						orange	orange	orange	orange	orange					
11						orange	black	black	black	orange					
10						light blue	light blue	light blue	light blue	light blue					
9		brown	brown	brown	light blue	light blue	light blue	black	light blue	light blue	light blue	brown	brown	brown	
8					light blue	light blue	light blue	light blue	light blue	light blue	light blue				
7					light blue	light blue	light blue	black	light blue	light blue	light blue				
6				light blue	light blue	light blue	light blue	light blue	light blue	light blue	light blue	light blue			
5				light blue	light blue	light blue	light blue	black	light blue	light blue	light blue	light blue			
4				light blue	light blue	light blue	light blue	light blue	light blue	light blue	light blue	light blue			
3				light blue	light blue	light blue	light blue	black	light blue	light blue	light blue	light blue			
2				light blue	light blue	light blue	light blue	light blue	light blue	light blue	light blue	light blue			
1				light blue	light blue	light blue	light blue	light blue	light blue	light blue	light blue	light blue			

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.